

# EXERCISE SOLUTIONS TO *THE RISING SEA: FOUNDATIONS OF ALGEBRAIC GEOMETRY*

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## 1. JUST ENOUGH CATEGORY THEORY TO BE DANGEROUS

### 1.1. Categories and functors.

**Exercise (1.1.A).** *Groupoids.*

- (1) *Make sense of the statement “a group is a groupoid with one element.”*

*Solution.* Let  $\mathcal{G}$  be a one element groupoid, call the single element  $x$ . Then by definition the identity morphism is in  $\text{hom}_{\mathcal{G}}(x, x)$ . We will take this to be the identity element of the group  $\text{hom}_{\mathcal{G}}(x, x)$ . Any two morphisms are composable, which gives the group composition, and by definition of groupoid any morphism is invertible, yielding the group inverse.

Conversely, if we have a group  $G$  we can make a category with one object whose morphisms are the elements of  $G$ , and whose identity, composition, and inverses are given by the group structure. This yields a groupoid.  $\square$

- (2) *Give an example of a groupoid that is not a group.*

*Solution.* A category with more than one object, and only the identity morphisms.  $\square$

**Exercise (1.1.B).** *If  $A$  is an object in a category  $\mathcal{C}$ , show that the invertible elements of  $\text{hom}_{\mathcal{C}}(A, A)$  form a group, called the automorphism group. What are the automorphism groups in  $\text{Set}$  and  $\text{Vect}_k$ ?*

*Proof.* This is exactly the same as part 1 above. The identity morphism serves as the identity, the inverse morphisms yield the group inverses, and the category axioms yield composition.

The automorphism groups in  $\text{Set}$  are the symmetric groups, and the automorphism groups in  $\text{Vect}_k$  are the general linear groups.  $\square$

**Exercise (1.1.C.).** *Fix a field  $k$ . Let  $(-)^{\vee\vee}$  be the double dual functor from the category of finite-dimensional  $k$ -vector spaces to itself. Show that  $(-)^{\vee\vee}$  is naturally isomorphic to the identity functor.*

*Proof.* For a fixed vector space  $V$ , define the map  $\alpha_V : V \rightarrow V^{\vee\vee}$  by  $v \mapsto (\phi \mapsto \phi(v))$ . This is injective because if  $\phi \mapsto \phi(v)$  is the zero map for every  $\phi$ , then  $v = 0$ , so by finite-dimensionality  $\alpha_V$  is an

isomorphism. And it remains to check that for any map of vector spaces  $f : V \rightarrow W$  and  $v \in V$ , that

$$\begin{aligned}\alpha_W(f(v)) &= (\phi \mapsto \phi(f(v))) \\ &= (\psi \mapsto \psi(v)) \circ f^\vee \\ &= f^{\vee\vee}(\psi \mapsto \psi(v)) \\ &= f^{\vee\vee}(\alpha_V(v))\end{aligned}$$

which completes the proof.  $\square$

**Exercise (1.1.D).** Let  $\mathcal{V}$  be the category whose objects are the  $k$ -vector spaces  $k^n$  for  $n \geq 0$ , and whose morphisms are linear transformations. There is a functor  $F$  from  $\mathcal{V}$  to the category of f.d.  $k$ -vector spaces sending each vector space to itself. Show that  $F$  is an equivalence of categories.

*Proof.* Following Vakil's remark about ignoring set-theoretic conditions, choose a basis for each finite dimensional  $k$ -vector space. For any vector space of the form  $k^n$ , choose the standard basis  $\{e_i\}$ .

Then define a functor  $G$  from f.d.  $k$ -vector spaces to  $\mathcal{V}$  by sending an  $n$ -dimensional  $k$ -vector space to  $k^n$ . For a morphism  $\phi$  from  $V$  to an  $m$ -dimensional vector space  $W$ , let  $\{v_1, \dots, v_n\}$  be the chosen basis for  $V$  and  $\{w_1, \dots, w_m\}$  the chosen basis for  $W$ . Then define  $G\phi$  to be the matrix whose  $(i, j)$ th entry is the coefficient of  $w_j$  in  $\phi(v_i)$ .

Define a natural transformation  $\alpha$  from  $FG$  to the identity functor where  $\alpha_V : FGV \rightarrow V$  sends  $FGV = k^n$  to  $V$  by mapping the standard basis vector  $e_i$  of  $FGV$  to the chosen  $i$ th basis vector  $v_i$  of  $V$ . Then  $\alpha_V$  is an isomorphism. For naturality, we have for a map  $\phi : V \rightarrow W$  that

$$\begin{aligned}\phi(\alpha_V(e_i)) &= \phi(v_i) \\ &= \sum_j (G\phi)_{i,j} w_j \\ &= \alpha_W\left(\sum_j (G\phi)_{i,j} e_j\right) \\ &= \alpha_W(FG\phi)(e_i).\end{aligned}$$

In the other direction,  $GF$  is the identity functor, which completes the proof.  $\square$

## 1.2. Universal properties determine an object up to unique isomorphism.

**Exercise (1.2.A).** Show that any two initial objects are uniquely isomorphic, and any two final objects are uniquely isomorphic.

*Proof.* Let  $i_1, i_2$  be initial. Then there is exactly one map  $f_{12} : i_1 \rightarrow i_2$  and likewise  $f_{21} : i_2 \rightarrow i_1$ . Since there is also exactly one map from  $i_1$  to itself, the identity, so because  $f_{21} \circ f_{12}$  is one such map, it must be the identity. Likewise  $f_{12} \circ f_{21}$  is the identity, which yields the unique isomorphisms. The argument is exactly the same for final objects.  $\square$

**Exercise (1.2.B).** What are the initial and final objects (if they exist) in Ring, Set, and Top, or the category of subsets of a set under inclusion, or the category of open subsets of a topological space under inclusion?

*Solution.* In Ring, the initial object is the integers, since 1 mapping to 1 dictates where the remaining integers go; the terminal object is the zero ring, since everything must map to zero. In Set, the initial object is the empty set; the final object is the singleton. In Top, the initial object is the empty space; the final object is the singleton. In the category of subsets or open sets, the initial object is the empty set/space; the terminal object is the entire set/space.  $\square$

**Exercise (1.2.C).** Let  $A$  be a commutative ring and  $S$  a multiplicative subset of  $A$ . Show that  $f : A \mapsto S^{-1}A$  via  $a \mapsto a/1$  is injective if and only if  $S$  contains no zerodivisors.

*Proof.* First suppose  $f$  is not injective, i.e. there is a nonzero element  $a \in A$  such that  $a/1 = 0/1$ . So there exists some  $s \in S$  such that  $s(a1 - 01) = sa = 0$ . Then  $s$  is a zerodivisor.

Conversely, suppose  $S$  contains a zerodivisor  $s$ , with  $st = 0$  for  $t$  nonzero. Then  $0 = st = s(t1 - 01)$ ,  $t/1 = 0/1 = 0$ , so  $f(t) = t/1 = 0/1$  and  $f$  fails to be injective.  $\square$

**Exercise (1.2.D).** Verify that  $\iota : A \rightarrow S^{-1}A$  is initial among  $A$ -algebras  $B$  where every element of  $S$  is sent to an invertible element in  $B$ , i.e. any map  $\phi : A \rightarrow B$  sending all elements of  $S$  to invertible elements in  $B$  factors through  $S^{-1}A$ .

*Proof.* Define  $\tilde{\phi} : S^{-1}A \rightarrow B$  via  $\tilde{\phi}(a/s) := \phi(a)\phi(s)^{-1}$ . This makes sense since we have assumed that  $\phi(s)$  is invertible. Let  $a, b \in A$  and  $s, t \in S$ . Then

$$\tilde{\phi}(ab/st) = \phi(ab)\phi(st)^{-1} = \phi(a)\phi(b)\phi(s)^{-1}\phi(t)^{-1} = \tilde{\phi}(a/s)\tilde{\phi}(b/t)$$

and

$$\begin{aligned} \tilde{\phi}(a/s + b/t) &= \tilde{\phi}((at + sb)/st) \\ &= \phi(at + sb)\phi(st)^{-1} \\ &= \phi(a)\phi(t)/\phi(s)\phi(t) + \phi(s)\phi(b)/\phi(s)\phi(t) \\ &= \tilde{\phi}(a/s) + \tilde{\phi}(b/t). \end{aligned}$$

so  $\tilde{\phi}$  is a ring homomorphism. And  $\tilde{\phi}(\iota(a)) = \tilde{\phi}(a/1) = \phi(a)/\phi(1) = \phi(a)$  as desired.  $\square$

**Exercise (1.2.E).** The localization of an  $A$ -module  $M$  defined via the universal property, actually exists.

*Proof.* Following the hint, define  $S^{-1}M$  to have elements of the form  $m/s$  where  $m \in M$  and  $s \in S$ , subject to the equivalence relation where  $m/s \sim m'/t$  if there exists an element  $s' \in S$  such that  $s'(ms - m't) = 0$ , with the standard operation and  $S^{-1}A$ -module operation. Define  $\iota : M \rightarrow S^{-1}M$  via  $m \mapsto m/1$ .

Let  $\phi : M \rightarrow N$  be an  $A$ -module map such that multiplication by any element of  $s$  is an isomorphism  $N \rightarrow N$ . Then we can define a map  $\tilde{\phi} : S^{-1}M \rightarrow N$  via  $m/s \mapsto s^{-1}\phi(m)$ , which makes sense taking  $s^{-1}$  to be the inverse to multiplication by  $s$ . This is an  $A$ -module map and  $\tilde{\phi}(\iota(m)) = \tilde{\phi}(m/1) = 1^{-1}\phi(m) = \phi(m)$ .  $\square$

**Exercise (1.2.F).** Properties of localization.

- (1) Localization commutes with finite products, i.e. for  $A$ -modules  $M_1, \dots, M_n$  provide an  $A$ -module isomorphism  $S^{-1}(M_1 \times \dots \times M_n) \rightarrow S^{-1}M_1 \times \dots \times S^{-1}M_n$ .

*Proof.* Define our map via  $(m_1, \dots, m_n)/s \mapsto (m_1/s, \dots, m_n/s)$ . For surjectivity, given an element  $(m_1/s_1, \dots, m_n/s_n)$ , the element

$$(s_2 \cdots s_n m_1, \dots, s_1 \cdots s_{n-1} m_n) / s_1 \cdots s_n$$

will map to it. For injectivity, suppose  $(m_1/s, \dots, m_n/s)$  is zero, i.e. there exist elements  $t_i \in S$  such that  $t_i m_i = 0$  for all  $i$ . Then  $t_1 \cdots t_n (m_1, \dots, m_n) = 0$ , so  $(m_1, \dots, m_n)/s = 0$ .  $\square$

- (2) Localization commutes with arbitrary direct sums.

*Proof.* Let  $\Lambda$  be an indexing set. Define a map  $S^{-1}(\bigoplus_{\lambda \in \Lambda} M_\lambda) \rightarrow \bigoplus_{\lambda \in \Lambda} S^{-1}M_\lambda$  via

$$(m_\lambda)_{\lambda \in \Lambda} / s \mapsto (m_\lambda / s)_{\lambda \in \Lambda}.$$

For surjectivity, given an element  $(m_\lambda / s)_\lambda$ , we know that only finitely many indices are nonzero, say  $\lambda_1, \dots, \lambda_k$ . Then let  $s = \lambda_1 \cdots \lambda_k$ , and consider the element  $(k_\lambda m_\lambda)_{\lambda \in \Lambda}$  where  $k_{\lambda_i} = s / \lambda_i$  for the nonzero indices and zero otherwise. Then it maps to what we want. For injectivity, suppose  $(m_\lambda / s)_{\lambda \in \Lambda}$  is zero, i.e. there exist elements  $t_\lambda$  such that  $t_\lambda m_\lambda = 0$  for all  $\lambda$ . Again only finitely many of the  $m_\lambda$  are nonzero, so take the corresponding  $t_\lambda$  and let their product be  $t$ . Then  $t(m_\lambda)_{\lambda \in \Lambda} = 0$  and we are done.  $\square$

- (3) *Localization does not commute with infinite products, i.e. the same map defined as above is not always an isomorphism for infinite products.*

*Proof.* We know that  $\mathbf{Z}$  localized at  $\mathbf{Z} \setminus \{0\}$  is  $\mathbf{Q}$ , so that  $\prod_{\mathbf{N}} \mathbf{Q}$  contains for example the element  $x := (1, 1/2, 1/3, \dots)$ . On the other hand, suppose there were an element in  $(\mathbf{Z} \setminus \{0\})^{-1}(\mathbf{Z}^{\mathbf{N}})$  mapping to  $x$ , i.e. some element  $(a_1, a_2, \dots) / s$  where  $a_i \in \mathbf{Z}$  and  $s \in \mathbf{Z} \setminus \{0\}$  such that  $a_n / s = 1/n$  for all  $n \in \mathbf{N}$ . Then  $na_n = s$  for all  $n \in \mathbf{N}$ , i.e.  $s$  is an integer divisible by all natural numbers, a contradiction.  $\square$

**Exercise (1.2.G).** Show that  $\mathbf{Z}/10 \otimes \mathbf{Z}/12 \cong \mathbf{Z}/2$ .

*Proof.* We define a map  $\tilde{f} : \mathbf{Z}/10 \times \mathbf{Z}/12 \rightarrow \mathbf{Z}/2$  via  $([a], [b]) \mapsto [ab]$ . This is clearly bilinear. If  $a = a' + 10n$  and  $b = b' + 12m$ , so  $[ab] = [a'b' + 12ma' + 10nb' + 120nm] = [a'b']$  in  $\mathbf{Z}/2$ , so this map is well defined. By the universal property of tensor products,  $\tilde{f}$  factors through the tensor product to yield a map  $f : \mathbf{Z}/10 \otimes \mathbf{Z}/12 \rightarrow \mathbf{Z}/2$  such that  $[a] \otimes [b] \mapsto [ab]$ .

Consider the map  $g : \mathbf{Z}/2 \rightarrow \mathbf{Z}/10 \otimes \mathbf{Z}/12$  given by  $[a] \mapsto [a] \otimes [1]$ . This is again bilinear, and if  $a = a' + 2n$ , then  $[a] \otimes [1] = [a' + 2n] \otimes [1] = [a'] \otimes [1] + [2n] \otimes [1] = [a'] \otimes [1] + [1] \otimes [12n] - [10n] \otimes [1] = [a']$ , so this is well defined.

We moreover have  $fg([a]) = f([a] \otimes [1]) = [a]$  and  $gf([a], [b]) = g([ab]) = [ab] \otimes [1] = [a] \otimes [b]$ . Hence  $f$  is an isomorphism. (Note that this argument works to show in general that  $\mathbf{Z}/m \otimes \mathbf{Z}/n \cong \mathbf{Z}/\gcd(m, n)$ .)  $\square$

**Exercise (1.2.H).** Show that  $(-) \otimes_A N$  is a covariant functor  $\text{Mod}_A \rightarrow \text{Mod}_A$ , and that it is right exact.

*Proof.* Let  $f : M \rightarrow M'$  be an  $A$ -module map. Then  $(m, n) \mapsto f(m) \otimes n$  is bilinear, so it yields a map  $f_* : M \otimes N \rightarrow M' \otimes N$ , which on generators is given by  $m \otimes n \mapsto f(m) \otimes n$ . This respects composition, thus making  $(-) \otimes N$  into a functor.

Let  $M \xrightarrow{f} M' \xrightarrow{g} M'' \xrightarrow{h} 0$  be exact. Tensoring, we get a complex

$$M \otimes N \xrightarrow{f_*} M' \otimes N \xrightarrow{g_*} M'' \otimes N \xrightarrow{h_*} 0,$$

which we need to check is exact. We know by the first isomorphism theorem that  $g$  induces an isomorphism  $M' / \text{im}(f) \rightarrow M''$ .

**Claim 1.1.** Tensor products commute with quotients.

*Proof.* Let  $P, Q, R$  be  $A$ -modules. Suppose  $Q$  is a submodule of  $P$ . Consider the map  $\tilde{f} : (P/Q) \times R \rightarrow (P \otimes R) / (Q \otimes R)$  given by  $([p], r) \mapsto [p \otimes r]$ . If  $[p] = [p']$ , then  $[p \otimes r] - [p' \otimes r] = [p - p' \otimes r] = [q \otimes r] = 0$  so this map is well defined. One can also check bilinearity. Hence it extends to a map  $f : (P/Q) \otimes R \rightarrow (P \otimes R) / (Q \otimes R)$ .

We now define a map  $\tilde{g} : P \times R \rightarrow (P/Q) \otimes R$  via  $(p, r) \mapsto [p] \otimes r$ . This is bilinear, hence yields a map  $g' : P \otimes R \rightarrow (P/Q) \otimes R$ , which further factors through the quotient by  $Q \otimes R$ , yielding a map  $g : (P \otimes R) / (Q \otimes R) \rightarrow (P/Q) \otimes R$ . And  $f, g$  are mutually inverse, yielding the claim.  $\square$

By the claim,  $g_*$  induces an isomorphism (since functors send isos to isos)

$$M'' \otimes N \leftarrow (M'/\text{im}(f)) \otimes N \cong (M' \otimes N)/\text{im}(f_*)$$

so  $\text{im}(f_*) = \ker(g_*)$ . And surjectivity of  $g_*$  follows directly from surjectivity of  $g$ .  $\square$

**Exercise (1.2.I).** *The tensor product  $(T, t)$  of two modules  $M, N$  is unique up to unique isomorphism. (Here  $t : M \times N \rightarrow T$  is the bilinear map given by the universal property.)*

*Proof.* Let  $(T', t')$  be another pair satisfying the universal property. Then  $t'$  must factor through  $T$ , i.e. there exists a unique map  $f : T \rightarrow T'$  such that  $f \circ t = t'$ . Likewise, there exists a unique map  $g : T' \rightarrow T$  such that  $g \circ t' = t$ . Plugging these two formulas into each other we get  $f \circ g \circ t' = t'$  and  $g \circ f \circ t = t$ .

Now we use the universal property again. Since  $t'$  is a bilinear map from  $M \times N$  to  $T'$ , it must factor uniquely through  $T'$ . One such factoring is as above,  $(f \circ g) \circ t' = t'$ , but another factoring is  $\text{id}_{T'} \circ t'$ , so by uniqueness,  $f \circ g = \text{id}_{T'}$ . Likewise,  $g \circ f = \text{id}_T$ , hence  $f, g$  are inverses, and the unique isomorphisms between  $T, T'$ .  $\square$

**Exercise (1.2.J).** *Show that the tensor product actually exists, i.e. the construction from §1.2.5 satisfies the universal property.*

*Proof.* Recall that  $M \otimes N$  is the free  $A$ -module generated by  $M \times N$ , quotiented by the following relations:

- (1)  $(m_1 + m_2, n) - (m_1, n) - (m_2, n)$
- (2)  $(m, n_1 + n_2) - (m, n_1) - (m, n_2)$
- (3)  $a(m, n) - (am, n)$
- (4)  $a(m, n) - (m, an)$

for  $a \in A, m, m_1, m_2 \in M, n, n_1, n_2 \in N$ . Let  $f : M \times N \rightarrow P$  be a bilinear  $A$ -module morphism. It follows by bilinearity that

- (1)  $f((m_1 + m_2, n) - (m_1, n) - (m_2, n)) = f(m_1, n) + f(m_2, n) - f(m_1, n) - f(m_2, n) = 0$
- (2)  $f((m, n_1 + n_2) - (m, n_1) - (m, n_2)) = f(m, n_1) + f(m, n_2) - f(m, n_1) - f(m, n_2) = 0$
- (3)  $f(a(m, n) - (am, n)) = af(m, n) - af(m, n) = 0$
- (4)  $f(a(m, n) - (m, an)) = af(m, n) - af(m, n) = 0$

so  $f$  factors uniquely through the quotient.  $\square$

**Exercise (1.2.K).** “Important exercise”

- (1) *If  $M$  is an  $A$ -module and  $\phi : A \rightarrow B$  is a ring morphism, give  $B \otimes_A M$  the structure of a  $B$ -module. Show that this gives a functor  $\text{Mod}_A \rightarrow \text{Mod}_B$ .*

*Proof.* Regard  $B$  as an  $A$ -module via the action  $A \times B \rightarrow B : (a, b) \mapsto \phi(a)b$ . Now  $B \otimes_A M$  makes sense as an  $A$ -module. Now for fixed  $b \in B$ , consider the map  $\tilde{f}_b : B \times M \rightarrow B \otimes_A M$  given by  $(b', m) \mapsto bb' \otimes m$ . We compute

$$\tilde{f}_b(\phi(a)b', m) = b\phi(a)b' \otimes m = \phi(a)(bb' \otimes m) = a \cdot \tilde{f}_b(b', m)$$

and

$$\tilde{f}_b(b', am) = bb' \otimes am = a \cdot (bb' \otimes m) = a \cdot \tilde{f}_b(b', m)$$

so  $\tilde{f}_b$  is  $A$ -bilinear, and it induces a map  $f_b : B \otimes_A M \rightarrow B \otimes_A M$ , which yields the  $B$ -module structure on  $B \otimes_A M$ .

To see that this is a functor, suppose we have an  $A$ -module morphism  $g : M \rightarrow N$ . Then we get a map  $B \times M \rightarrow B \otimes_A N$  given by  $(b, m) \mapsto b \otimes g(m)$ . This is again  $A$ -bilinear, so it induces a map  $g_* : B \otimes_A M \rightarrow B \otimes_A N$ , which acts on elementary tensors via  $b \otimes m \mapsto b \otimes g(m)$ .  $\square$

(2) Following notation above, if  $\psi : A \rightarrow C$  is a ring morphism, then  $B \otimes_A C$  has a ring structure.

*Proof.* Again,  $C$  becomes an  $A$ -module via the action  $(a, c) \mapsto \psi(a)c$ . Then  $B \otimes_A C$  automatically gets an abelian group structure. To define the ring product, note that specifying a map

$$(B \otimes C) \times (B \otimes C) \rightarrow B \otimes C$$

is the same thing as specifying a map

$$B \otimes C \rightarrow \text{hom}_{\text{Mod}_A}(B \otimes C, B \otimes C)$$

so we can start here. First define a map

$$B \times C \rightarrow \text{hom}_{\text{Mod}_A}(B \times C, B \otimes C) : (b_1, c_1) \mapsto ((b_2, c_2) \mapsto b_1 b_2 \otimes c_1 c_2).$$

By properties of the tensor product, the image of this map is an  $A$ -bilinear map, so we get a map

$$B \times C \rightarrow \text{hom}_{\text{Mod}_A}(B \otimes C, B \otimes C) : b_1, c_1 \mapsto (b_2 \otimes c_2 \mapsto b_1 b_2 \otimes c_1 c_2).$$

Again, this is also  $A$ -bilinear. For example we can check

$$\phi(a)b_1 \otimes c_1 \mapsto (b_2 \otimes c_2 \mapsto \phi(a)b_1 b_2 \otimes c_1 c_2) = (b_2 \otimes c_2 \mapsto \phi(a)(b_1 b_2 \otimes c_1 c_2)).$$

Hence we get a map

$$B \otimes C \rightarrow \text{hom}_{\text{Mod}_A}(B \otimes C, B \otimes C) : b_1 \otimes c_1 \mapsto (b_2 \otimes c_2 \mapsto b_1 b_2 \otimes c_1 c_2)$$

which is equivalent to the map

$$\mu : (B \otimes C) \times (B \otimes C) \rightarrow B \otimes C : (b_1 \otimes c_1), (b_2 \otimes c_2) \mapsto b_1 b_2 \otimes c_1 c_2$$

which is  $A$ -bilinear by definition since it comes from the universal property. Now for  $x, y, z \in B \otimes C$ , we have by bilinearity

$$\mu(x + y, z) = \mu(x, z) + \mu(y, z), \quad \mu(x, y + z) = \mu(x, y) + \mu(x, z)$$

so this satisfies distributivity. The identity element is  $1_B \otimes 1_C$ , and  $\mu$  is obviously associative, so we are done.  $\square$

**Exercise (1.2.L).** If  $S$  is a multiplicative subset of  $A$  and  $M$  is an  $A$ -module, describe a natural isomorphism  $(S^{-1}A) \otimes_A M \rightarrow S^{-1}M$  (as both  $A$ -modules and  $S^{-1}A$ -modules).

*Proof.* Consider first the map  $S^{-1}A \times M \rightarrow S^{-1}M$  given by  $(a/s, m) \mapsto (a \cdot m)/s$ . This is  $A$ -bilinear since

$$(a' \cdot a/s, m) = (a' a/s, m) \mapsto (aa' \cdot m)/s = a \cdot (a' \cdot m)/s$$

and

$$(a/s, a' \cdot m) \mapsto (a \cdot (a' \cdot m))/s = (aa' \cdot m)/s = a \cdot (a' \cdot m)/s.$$

Hence this induces a map  $f : (S^{-1}A) \otimes_A M \rightarrow S^{-1}M$  which acts on elementary tensors as

$$f(a/s \otimes m) = (a \cdot m)/s.$$

We can define an inverse map as follows. First consider the map  $\tilde{g} : M \rightarrow (S^{-1}A) \otimes_A M$  via  $m \mapsto 1/1 \otimes m$ . Since multiplication by elements of  $S$  are invertible in  $(S^{-1}A) \otimes_A M$ , the map  $\tilde{g}$  factors through the localization to yield a map  $g : S^{-1}M \rightarrow (S^{-1}A) \otimes_A M$  which acts via  $m/s \mapsto s^{-1}(1/1 \otimes m) = (1/s \otimes m)$ . Then we can check

$$gf(a/s \otimes m) = g((a \cdot m)/s) = (1/s \otimes a \cdot m) = (a/s \otimes m)$$

and

$$fg(m/s) = f(1/s \otimes m) = m/s.$$

It remains to check that  $f$  is also a  $S^{-1}A$ -module map. Recall from Exercise 1.2.K.1 that  $S^{-1}A \otimes_A M$  has the structure of a  $S^{-1}A$ -module via the map  $A \rightarrow S^{-1}A, a \mapsto a/1$ . So for  $a/s, a'/s' \in S^{-1}A, m \in M$ , we have

$$f(a/s \cdot (a'/s') \otimes m) = f(aa'/ss' \otimes m) = (aa' \cdot m)/ss' = (a'/s') \cdot (am/s)$$

as desired. For naturality, we need to check that the following diagram commutes for any map of  $A$ -modules  $\phi : M \rightarrow N$ :

$$\begin{array}{ccc} (S^{-1}A) \otimes_A M & \xrightarrow{f} & S^{-1}M \\ \phi_* \downarrow & & \downarrow \phi_* \\ (S^{-1}A) \otimes_A N & \xrightarrow{f} & S^{-1}N \end{array}$$

but this is not too bad, for following the top and right arrows we have

$$\phi_*(f(a/s \otimes m)) = \phi_*((a \cdot m)/s) = \phi(a \cdot m)/s$$

and following the left and bottom ones we have

$$f(\phi_*(a/s \otimes m)) = f(a/s \otimes \phi(m)) = (a \cdot \phi(m))/s = \phi(a \cdot m)/s$$

as desired.  $\square$

**Exercise (1.2.M).** Show that tensor products commute with arbitrary direct sums, i.e. for  $A$ -modules  $M$  and  $\{N_\alpha\}_{\alpha \in \Lambda}$ , define an isomorphism  $M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha) \rightarrow \oplus_{\alpha \in \Lambda} (M \otimes N_\alpha)$ .

*Proof.* As with every map out of a tensor product, we begin by defining a map

$$\tilde{f} : M \times (\oplus_{\alpha \in \Lambda} N_\alpha) \rightarrow \oplus_{\alpha \in \Lambda} (M \otimes N_\alpha), (m, (n_\alpha)_{\alpha \in \Lambda}) \mapsto (m \otimes n_\alpha)_{\alpha \in \Lambda}.$$

This, being bilinear, induces a map  $f : M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha) \rightarrow \oplus_{\alpha \in \Lambda} (M \otimes N_\alpha)$  which acts on elementary tensors via  $m \otimes (n_\alpha)_{\alpha \in \Lambda} \mapsto (m \otimes n_\alpha)_{\alpha \in \Lambda}$ .

For each  $\alpha \in \Lambda$ , consider the map

$$\tilde{g}_\alpha : M \times N_\alpha \rightarrow M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha), (m, n) \mapsto m \otimes \iota_\alpha(n)$$

where  $\iota_\alpha : N_\alpha \rightarrow \oplus_{\alpha \in \Lambda} N_\alpha$  is the inclusion. This is  $A$ -bilinear, so it induces a map  $g_\alpha : M \otimes N_\alpha \rightarrow M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha)$ . The sum of all the  $g_\alpha$  yields a map  $g := \oplus_{\alpha \in \Lambda} (M \otimes N_\alpha) \rightarrow M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha)$ .

And we check

$$\begin{aligned} fg((m_\alpha \otimes n_\alpha)_{\alpha \in \Lambda}) &= f \sum_{\alpha \in \Lambda} (m_\alpha \otimes \iota_\alpha(n_\alpha)) \\ &= \sum_{\alpha \in \Lambda} f(m_\alpha \otimes \iota_\alpha(n_\alpha)) \\ &= \sum_{\alpha \in \Lambda} (\dots, 0, m_\alpha \otimes n_\alpha, 0, \dots) \\ &= (m_\alpha \otimes n_\alpha)_{\alpha \in \Lambda} \end{aligned}$$

and

$$gf(m \otimes (n_\alpha)_{\alpha \in \Lambda}) = g((m \otimes n_\alpha)_{\alpha \in \Lambda}) = \sum_{\alpha \in \Lambda} m \otimes \iota_\alpha(n_\alpha) = m \otimes \sum_{\alpha \in \Lambda} \iota_\alpha(n_\alpha) = m \otimes (n_\alpha)_{\alpha \in \Lambda}$$

so  $f$  is an isomorphism.  $\square$

**Exercise (1.2.H).** Show that in  $\text{Set}$ , the fiber product is given by  $X \times_Z Y = \{(x, y) \in X \times Y : \alpha(x) = \beta(y)\}$ .

*Proof.* Call the right hand side of above  $R$ . Let  $W$  be a set with maps  $f_X : W \rightarrow X$  and  $f_Y : W \rightarrow Y$  such that  $\alpha \circ f_X = \beta \circ f_Y$ . Define the map  $\phi : W \rightarrow R$  via  $\phi(w) = (f_X(w), f_Y(w))$ . This actually lands in  $R$  since  $\alpha \circ \pi_X(\phi(w)) = \alpha \circ f_X(w) = \beta \circ f_Y(w) = \beta \circ \pi_Y(\phi(w))$ . Then clearly  $\pi_X \circ \phi = f_X$ , and  $\pi_Y \circ \phi = f_Y$ . Thus  $R$  satisfies the universal property so it is the fiber product in the category of sets.  $\square$

**Exercise (1.2.O).** If  $X$  is a topological space, show that fiber products always exist in the category of open sets of  $X$  by describing what a fibered product is.

*Proof.* Remember that the morphisms here are just inclusions of open sets. So suppose we have open sets  $U, U'$  mapping into  $W$ , i.e.  $U, U' \subset W$ . We claim that the fiber product here is just  $U \cap U'$  (intersections of two open sets are open), with the morphisms given by inclusions into  $U, U'$ . To see this, given another open set  $V$  with morphisms  $V \rightarrow U$  and  $V \rightarrow U'$  that agree with  $U \rightarrow W$  and  $U' \rightarrow W$ , then we know that  $V$  must be a subset of both  $U$  and  $U'$ . It follows that  $V$  is a subset of  $U \cap U'$ . And by definition, all the morphisms are inclusions, so everything commutes.  $\square$

**Exercise (1.2.P).** If  $Z$  is the final object in the category  $\mathcal{C}$ , show that the fiber product  $X \times_Z Y$  is given by the product  $X \times Y$ . (Assume all (fiber) products exist.)

*Proof.* Let  $W \in \mathcal{C}$  with maps  $f_X : W \rightarrow X$  and  $f_Y : W \rightarrow Y$  as in the setup of the universal property. Then by the universal property of products, we get a map  $\phi : W \rightarrow X \times Y$  which factors  $f_X$  and  $f_Y$ . Moreover, since  $Z$  is the final object, the compositions  $X \times Y \rightarrow X \rightarrow Z$  and  $X \times Y \rightarrow Y \rightarrow Z$  must agree.  $\square$

**Exercise (1.2.Q).** Stacking Cartesian squares gives a Cartesian rectangle:

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ W & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Z \end{array}$$

*Proof.* Suppose we have maps  $f_Y : A \rightarrow Y$  and  $f_V : A \rightarrow V$  such that their postcompositions with  $Y \rightarrow Z$  and  $V \rightarrow Z$  agree. Then since the bottom square is Cartesian, there is a unique map  $f_{AW} : A \rightarrow W$  which factors  $f_Y$  and  $f_{VX} \circ f_V$ , i.e.  $f_{WX} \circ f_{AW} = f_{VX} \circ f_V$ . Now since the top square is Cartesian, there is a unique map  $f_{AU} : A \rightarrow U$  factoring  $f_{AW}$  (hence  $f_Y$ ) and  $f_V$ , completing the proof. See below for a figure of all the maps involved:

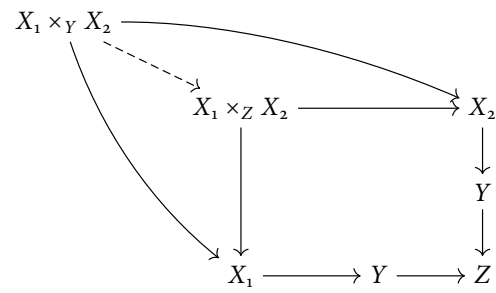
$$\begin{array}{ccccc} & & & & \\ & & & & \\ & & & & \\ A & \xrightarrow{f_V} & V & & \\ & \searrow f_{AU} & \downarrow & & \\ & & U & \longrightarrow & V \\ & \searrow f_{AW} & \downarrow & & \downarrow f_{VX} \\ & & W & \xrightarrow{f_{WX}} & X \\ & \searrow f_Y & \downarrow & & \downarrow \\ & & Y & \longrightarrow & Z \end{array}$$

$\square$

**Exercise (1.2.R).** Given morphisms  $X_1 \rightarrow Y$ ,  $X_2 \rightarrow Y$ , and  $Y \rightarrow Z$ , there is a natural morphism  $X_1 \times_Y X_2 \rightarrow X_1 \times_Z X_2$  (assuming both fiber products exist).



*Proof.* Consider the following diagram:



where the outside arrows commute by definition. Hence by the universal property we get the dotted morphism.  $\square$